

Triangle Design Studio

Unit 5: Architecture Firm • Lesson 5-3 • Flagship • EduWonderLab

UNIT 5 • LESSON 5-3 ★ FLAGSHIP • TEACHER NOTES

Standard 6.G.1

FORMAT Design + justify project	TIME 30–35 min
GROUPING Pairs or teams	TOPIC Area of Triangles (Flagship)
STANDARD 6.G.1	PREP LEVEL Medium — print grid paper, triangle templates, design brief

Learning goal *I can design a banner with triangles, find its total area, and justify the base-height pairs I used.*

Teacher setup

- Print grid paper and templates per team. Each team designs a pennant or banner from triangles.
- Students calculate the total area and justify each base-height pair.
- Extension: two triangles with equal area but different shapes.

Materials

Grid paper, Triangle templates, Design constraints card, Pencil.

Differentiation & language support

- Grid paper makes base and height easy to count (SPED).
- Sentence stem: “I used base ____ and height ____ because ____.”
- ESOL: provide a finished example banner to model the task.

Key vocabulary (English / Español):

Term	Español	What it means
design	diseño	a plan for how something looks
total area	área total	the sum of all the areas

Quick teacher check: *Check the height found on Card 4 ($h = 8$).*

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Name: _____ Date: _____ Class: _____

Your goal: *I can design a banner with triangles, find its total area, and justify the base-height pairs I used.*

What you need

Grid paper, Triangle templates, Design constraints card.

How to play

1. Read the design task and sketch your triangles on grid paper.
2. Find the area of each triangle with $A = \frac{1}{2} \times \text{base} \times \text{height}$.
3. Add the areas for the total.
4. Justify which base and height you used for each triangle.

Design Tasks

Design a banner from triangles, find the area of each, and add for the total.

1. A pennant triangle has base 12, height 8. Area?	2. A banner uses 3 identical triangles, each base 6, height 10. Total area?
3. Show two triangles with equal area: base 8 height 6, and base 12 height 4.	4. A flag triangle has area 36 and base 9. Find the height.
5. A bunting has 4 triangles, each base 10, height 6. Total area?	6. Challenge: Design two triangles with area 30 but different shapes.

Design log (area of each triangle + total):

Explain your thinking

Explain how two triangles can look different but still have the same area.

Sentence frame: *I used base ___ and height ___ because ___.*

★ **Challenge:** *Design a banner with a total area of exactly 100 sq units using at least three triangles.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. A pennant triangle has base 12, height 8. Area? Answer: 48 sq units	2. A banner uses 3 identical triangles, each base 6, height 10. Total area? Answer: 90 sq units (3×30)
3. Show two triangles with equal area: base 8 height 6, and base 12 height 4. Answer: Both = 24 sq units	4. A flag triangle has area 36 and base 9. Find the height. Answer: 8 ($36 = \frac{1}{2} \times 9 \times h$)
5. A bunting has 4 triangles, each base 10, height 6. Total area? Answer: 120 sq units (4×30)	6. Challenge: Design two triangles with area 30 but different shapes. Answer: e.g., base 10 height 6, and base 5 height 12

Teaching notes & look-fors

- Card 4: $36 = \frac{1}{2} \times 9 \times h \rightarrow 72 = 9h \rightarrow h = 8$.
- Equal area with different shapes shows base and height can trade off.